

Research Paper

Zingerone effect against *Candida albicans* growth and biofilm production

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ABSTRACT

Background: The increasing resistance of *Candida albicans* biofilms underscores the urgent need for effective antifungals. This study evaluated the efficacy of zingerone and elucidated its mode of action against *C. albicans* ATCC 90028 and clinical isolate C1.

Experimental Procedure: Minimum inhibitory concentrations (MICs) of zingerone were determined using CLSI methods against planktonic cells, biofilm formation, and yeast-to-hyphal transition. The mode of action was investigated through fluorescent microscopy, ergosterol assays, cell cycle analysis, and RT-PCR for gene expression.

Key Results: Zingerone inhibited planktonic growth and biofilm formation at in *C. albicans* ATCC 90028 and clinical isolate C1 at 2 mg/mL 4 mg/mL and 1 mg/mL and 2 mg/mL respectively. Treatment with the MIC concentration caused significant cell cycle arrest at the G0/G1 phase, halting proliferation in both the strains. Propidium iodide Staining revealed compromised membrane integrity in both the strains. Also, acridine orange and ethidium bromide dual staining showed increased dead cell proportions in *C. albicans* ATCC 90028. RT-PCR studies showed downregulation of *BCY1*, *PDE2*, *EFG1*, and upregulation of negative regulators *NRG1*, *TUP1* disrupting growth and virulence pathways. Zingerone induced elevated reactive oxygen species (ROS) levels, triggering apoptosis, evidenced by DNA fragmentation and upregulation of apoptotic markers. It also inhibited ergosterol synthesis in a concentration-dependent manner, crucial for membrane integrity. Importantly, zingerone exhibited minimal hemolytic activity. In an *in vivo* silkworm model, zingerone demonstrated significant antifungal efficacy, protecting silkworms from infection. It also modulated stress response genes, highlighting its multifaceted action.

Conclusions: *In vitro* and *in vivo* findings confirm the potent antifungal efficacy of zingerone against *C. albicans* ATCC 90028 and clinical isolate C1, suggesting its promising potential as a therapeutic agent that warrants further exploration.

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Introduction

C. albicans is indeed a commensal organism commonly established on human skin, oral cavity, and within the alimentary canal [1]. It is a major opportunistic fungal pathogen responsible for a wide range of infections, from superficial mucosal infections to life-threatening systemic diseases [2]. Candidiasis, caused by *Candida albicans*, is a prevalent opportunistic fungal infection that poses a significant threat to immunocompromised individuals, such as those undergoing chemotherapy, organ transplants, or living with acquired immunodeficiency syndrome (AIDS). The global burden of candidiasis is

considerable, with systemic infections exhibiting mortality rates exceeding 40 %, even with the availability of antifungal treatments [2]. The increasing resistance of *C. albicans* to antifungal drugs, particularly in the case of invasive candidiasis, has led to prolonged hospital stays and elevated healthcare costs [3].

A key challenge in managing *C. albicans* infections is its ability to form biofilms, structured microbial communities that are highly resistant to antifungal therapies [4]. Biofilm formation on medical devices such as catheters and implants complicate treatment, as biofilm-associated infections are resistant to azoles and polyenes due to factors such as reduced drug penetration and altered metabolic states [5]. This resistance, combined with the emergence of multi-drug-resistant strains, underscores the need for new therapeutic strategies [6].

Despite the availability of drugs like fluconazole and Amphotericin B, these treatments have limitations. Fluconazole's efficacy

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Abbreviations

AO	Acridine orange
CLSI	Clinical standard institute
EDTA	Ethylene diamine tetraacetic acid
ER	Ethidium bromide
MFC	Minimum fungicidal concentration
MIC	Minimum inhibitory concentration for planktonic growth
PBS	Phosphate-buffered saline
PI	Propidium iodide
ROS	Reactive oxygen species
SEM	Scanning electron microscopy
XTT	(2, 3-Bis-(2-Methoxy-4-Nitro-5-Sulfophenyl)-2H-Tetrazolium-5-Carboxanilide)
YPD	Yeast Peptone Dextrose

against *C. albicans* biofilms is limited, and resistance to this drug is increasingly reported [7]. Although Amphotericin B is effective against systemic infections, its nephrotoxicity restricts its long-term use [8]. The urgent need for alternative antifungal agents has directed research towards natural products with broad-spectrum biological activities. In addition to *C. albicans*, other non-*albicans* species such as *Stephanoascus ciferrii* have been increasingly associated with infections, particularly in immunocompromised patients. Recent studies highlight the rising concern of antifungal resistance among these species, complicating treatment strategies [9]. Understanding the virulence mechanisms and resistance profiles of non-*albicans* species is crucial as they present a growing challenge in clinical settings. This context underscores the importance of exploring novel antifungal agents, such as zingerone, which may have broad-spectrum efficacy against multiple *Candida* species.

Zingerone, a phenolic compound derived from ginger (*Zingiber officinale*), has shown strong antifungal and anti-biofilm activities. Zingerone possesses anti-inflammatory, antiangiogenic, anticancer, and antioxidant properties [10,11] (Fig. 1). Zingerone have demonstrated potent antifungal activity against *C. albicans*, but the precise mechanisms of action are not fully understood. Zingerone has been shown to affect *C. albicans* membrane, but further studies are needed to elucidate these mechanisms fully. Zingerone disrupts biofilm formation, impairs membrane integrity [12].

The objective of this study is to assess the antifungal efficacy of zingerone against planktonic and biofilm forms of *C. albicans*, focusing on its impact on membrane integrity, cell cycle analysis, accumulation reactive oxygen species and gene expression. Also, an *in vivo* silkworm model was used to demonstrate antifungal efficacy of zingerone against *C. albicans*. By investigating these effects, this research aims to contribute to the development of safer, more effective antifungal treatments, especially for drug-resistant and biofilm-associated infections.

Research methodology

Strains and test molecules

Candida albicans ATCC 90028 was procured from IMTECH, Chandigarh and clinical isolate of *C. albicans* C1 was obtained from D. Y. Patil Hospital Kadamwadi, Kolhapur. Zingerone was obtained from Sigma Aldrich (India).

Media and chemicals

The strain was cultured on a Yeast-Peptone-Dextrose (YPD) agar slant and stored at 4°C. YPD medium was prepared by dissolving the components in distilled water, with the pH adjusted to 6.5. YPD agar

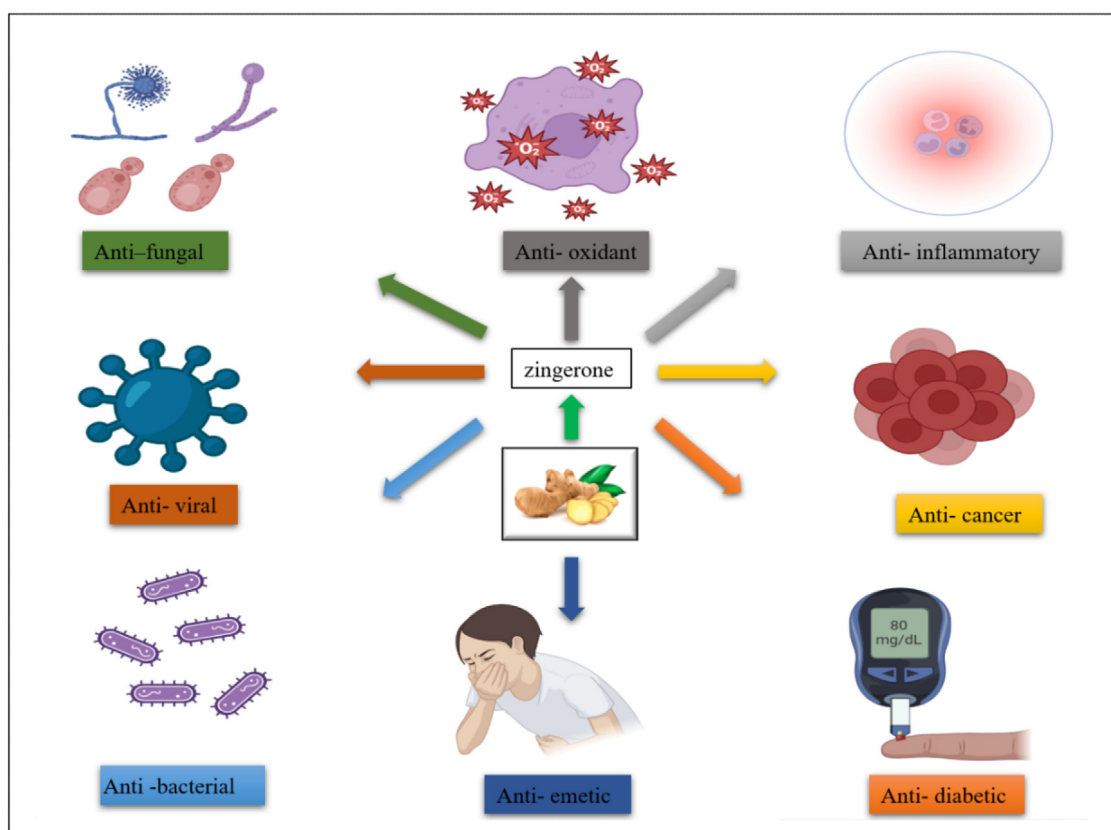


Fig. 1. Bioactive properties of Zingerone.

medium was made by adding 1.5 % agar to YPD broth. Before each assay, a colony from the plate was inoculated into 50 mL of YPD broth and incubated overnight at 30°C in an orbital shaker at 100 rpm. The cultured cells were harvested by centrifugation at 2000 rpm for 5 minutes and washed three times with sterile PBS to obtain a concentration of 1×10^9 cells/mL. A 20% fetal bovine serum (FBS) solution was prepared in distilled water. RPMI-1640 medium (with L-Glutamine, without sodium bicarbonate) was dissolved in distilled water and buffered with MOPS, with the pH adjusted to 7. The medium was sterilized using a 0.22 μ m syringe filter. The stock solution of zingerone was prepared in 2 % dimethyl sulfoxide (DMSO). The XTT reagent, (2,3-bis(2-methoxy-4-nitro-5-sulfophenyl)-2H-tetrazolium-5-carboxanilide), was purchased from SRL Pvt. Ltd. Maharashtra India (Catalog number 18681). All other media components, chemicals, and reagents were purchased from Hi-Media Laboratories Ltd., Mumbai, India. Amphotericin B was used as the standard control in this study.

Minimum inhibitory concentration (MIC)

The MIC of zingerone and amphotericin B on planktonic cells of *C. albicans* ATCC 90028 and C1 isolate was established through the utilization of the microbroth dilution method following the Standard CLSI M27-Ed4 guidelines. Different concentrations of zingerone, ranging from 0.125 mg/mL to 4 mg/mL, were made in RPMI 1640. The cell count utilized for the experiment was 1×10^3 cells/mL, and were prepared in RPMI 1640. 100 μ l of the prepared *C. albicans* cells were added in 96-well plate, Afterward, 100 μ l of varying concentrations of zingerone and amphotericin B were added. The plate was then incubated at 35°C, and after 48 h the growth was assessed by microplate reader (Multiskan Sky from Thermo Fisher Corp. USA) by quantifying the optical density at 620 nm [13].

Minimum fungicidal concentration (MFC)

To determine the minimum fungicidal concentration (MFC) against *C. albicans* ATCC 90028 and the C1 isolate, concentrations at or above the MIC were used. The cells were treated with zingerone and amphotericin B for 48 hours, then spread on sterile YPD agar plates. These plates were incubated at 30°C for 24 hours, after which they were checked for growth. The lowest concentration of zingerone that completely inhibited growth was recorded as the MFC [14].

Kill curve assay

Time required to kill *C. albicans* ATCC 90028 and C1 isolate cells by the treatment of zingerone was determined by time dependent kill curve assay method with modification described by [13]. 2.5×10^3 cells/mL were added to 10 mL of sterile YPD broth. The MFC concentration of zingerone was added to the medium. The flask was incubated at 30 °C. 1 mL of broth from the flask was taken at different time intervals (0, 15 min, 30 min, 1 h, 2 h, 4 h, 8 h, and 24 h). The cells were harvested by centrifugation washed twice with PBS. The 20 μ l of cell suspension was inoculated on YPD agar plate. The plates were incubated at 30 °C for 24 h. The number of colonies were counted compared with control without treatment of zingerone.

Germ tube test

Yeast to hyphal morphogenesis was studied using 96-well polystyrene plate assay. *C. albicans* ATCC90028 and C1 isolate cells were inoculated in 20 % fetal bovine serum prepared in distilled water to get 0.5 to 1×10^6 cells/ml and various concentrations of zingerone were added. Wells without any concentration of zingerone were kept as a control. Final volume of assay system in each well was kept at 200 μ l. The plates were incubated at 37°C at 100 rpm on an orbital

shaker for 90 min and cells were observed microscopically. Every time 100 cells were counted and the numbers of yeast and germ tube forms were noted. Percentage of germ tube formation in each well compared with control was calculated [15].

Metabolic activity assay

A microplate-based experiment was used to detect the influence of zingerone and amphotericin B on metabolic activity of *C. albicans* ATCC 90028 and C1 isolate. In order to prepare a series of zingerone and amphotericin B concentrations, PBS was used. Subsequently, 50 μ l aliquots of these concentrations were added into the wells. In each well, 50 μ l of 1×10^7 cells/mL were added. The microtiter plate was then kept at 37°C (90 min) in an incubator (REMI, India). By using the XTT assay relative metabolic activity was evaluated. Following the washes, XTT-menadione solution (100 μ l) was added. Subsequently, the plate was placed in an incubator at 37°C for a duration of 5 h. The quantification of the color produced by the water-soluble formazan product was carried out using a 96-well plate reader (Multiskan Sky by Thermo Fisher) at a wavelength of 490 nm [16].

Biofilm growth assay

Biofilm development was conducted on tissue culture-treated plate. To facilitate the adhesion of the cells, 100 μ l of 1×10^7 cells/mL of *C. albicans* ATCC 90028 and C1 isolate was added in the wells, the plate was kept at 37°C for 90 minutes at 100 rpm. PBS was used to remove non-adherent cells from the wells. Different concentrations of zingerone and amphotericin B were added into the wells, along with 100 μ l of RPMI-1640 media. The plate was kept for 48 h to form biofilm at 37°C. By employing the crystal violet assay the quantification of biofilm was performed [17].

Mature biofilm assay

The impact of zingerone and amphotericin B on mature biofilm structure of *C. albicans* ATCC 90028 and C1 isolate was assessed. Tissue culture-treated plates were employed for this purpose. Different concentrations of zingerone and amphotericin B, were made in RPMI 1640, and added into the plate containing mature biofilm of *C. albicans* ATCC 90028 and C1 isolate. Subsequently, By employing the crystal violet assay the quantification of biofilm was done [18].

Crystal violet assay

After incubation of 48-hour at 37°C, non-adherent cells were removed using sterile PBS. The remaining biofilms were then stained with 0.4% crystal violet for 20 minutes. Excess crystal violet was rinsed off three times with H₂O, and the stain was extracted using 95% ethanol. Absorbance was measured at 575 nm using a microplate reader (Multiskan Sky by Thermo Fisher) [19].

Microscopic imaging

After incubation, the wells were washed three times with sterile PBS. 0.4% of crystal violet solution was added to each well. After 20 min the solution was removed and well were washed with PBS. The plate was observed under inverted microscope and images were captured [19].

Scanning electron microscopy

Morphological analysis of *C. albicans* ATCC 90028 and C1 isolate biofilm was done by SEM. For the attachment of *C. albicans* ATCC 90028 and C1 isolate cells on catheter pieces, a procedure was followed. The cells were exposed to the catheters at 37°C for 90 minutes

with gentle agitation at 50 rpm. A cell suspension of 1×10^7 cells/mL was used for adhesion and MIC concentration of zingerone was added. The wells containing catheter without zingerone was served as control. After this step, the catheters were rinsed and placed in 2 mL of RPMI 1640 for the development of biofilm at 37°C for 24 h.

For the SEM (Scanning Electron Microscope), the samples were initially fixed in 2 % glutaraldehyde at 4°C for 24 h. After fixation, the samples were dried through different graded alcohols. Following dehydration, an automated gold coater was utilized to apply a gold coating to the samples. The samples were then mounted onto stubs. Using a scanning electron microscope, images of the samples were captured by JEOL 6360 (Tokyo, Japan) [20].

Confocal microscopy

Biofilm eradication was observed with confocal microscopy by Ahamad et al. [21] with slight modifications. Biofilm of *C. albicans* ATCC 90028 and C1 isolate was developed on the bottoms of 96 well plate in the presence of zingerone, with a control group representing the absence of zingerone. In summary, biofilms underwent treatment with zingerone at concentration of MIC. The samples were rinsed with PBS, stained with acridine orange dye the biofilm structures were visualized using a 40x oil immersion lens in confocal laser scanning microscopy (CLSM) imaging (Zeiss LSM710).

Ergosterol assay

A single colony of *C. albicans* ATCC 90028 and C1 isolate from a 24-hour-old SDA plate was inoculated into 50 mL of Sabourad broth. Different test concentrations of zingerone were added in each flask. Subsequently, all the flasks were kept in incubator at 30°C (16 h). Following the incubation period, the cells were centrifuged at 2,700 rpm (5 min). To each pellet, 25 % alcoholic KOH solution was introduced, and the mixture was vortexed for 1 minute. The cell suspensions were transferred to sterile tubes and placed in a water bath for 1 hour (80°C). The tubes were cooled, and to obtain sterols 1 mL of D/W and 3 mL of n-heptane were added. Before spectrophotometric analysis, the sterol extract was diluted in 100 % ethanol. An analysis was conducted within the wavelength range of 240 to 300 nm (Multiskan Sky by Thermo Fisher) [22].

Hemolytic activity

The hemolytic activity of RBCs was assessed through the procedure in which blood was obtained from a blood bank and collected in a tube containing EDTA to prevent clotting. The blood sample was then centrifuged. The resulting pellet of RBCs was suspended in PBS at a concentration of 10 % v/v. RBC solution was further diluted 1:10 in PBS. Aliquots from the diluted RBC suspension were combined with 100 μ l of zingerone at various concentrations, all in the same buffer, within Eppendorf tubes. For complete haemolysis, 1 % Triton X-100 was employed. After an incubation (37°C), the content was centrifuged at 2000 rpm (20°C). Following centrifugation, 150 μ l of the supernatant transferred to a Microtiter plate. The optical density (OD) was recorded (450 nm). The calculation of the percentage of hemolysis was performed using formula: % of haemolysis = [(A 450 of zingerone treated Sample – A 450 of saline treated sample) / (A 450 of 1 % Triton X treated sample – A 450 of Saline treated sample)] $\times$ 100 [17].

Cell cycle analysis

Log-phase cells of *C. albicans* ATCC 90028 and C1 isolate were collected from a 24 h old culture. The cells were rinsed using PBS, and then a concentration of 2×10^7 cells/mL was introduced into 10 mL of RPMI 1640 having a MIC concentration of zingerone (2 mg/mL and

4 mg/mL). RPMI 1640 without the drug was used as the control in the experiment. The flask was placed in an incubator at 30°C for 4 hours. The cells were obtained using a centrifuge at 6000 rpm (3 min). Following centrifugation, the cells washed with chilled PBS and then fixed by using chilled 70 % alcohol. Subsequently, RNase treatment was administered to the cells, and they were stained with propidium iodide before being analyzed using FACS (FACS Diva Version 6.1.3) [18].

Assessment of reactive oxygen species produced intracellularly

The concentration of ROS was quantified employing H₂DCF (2', 7'-Dichlorodihydrofluorescein diacetate). The *C. albicans* ATCC 90028 and C1 isolate cells were exposed to the MIC concentration of zingerone (test) as well as to H₂O₂ (hydrogen peroxide, positive control). The cells without any treatment were served as negative control. Following 4 hours of incubation (30°C), the cells were collected and then rinsed with sterile PBS (Phosphate-Buffered Saline). Afterwards, 40 μ g/mL of H₂DCF was introduced, and the placed in the darkness for 30 minutes. Following incubation, the cells were rinsed with sterile PBS. The fluorescence intensity was assessed using a spectrofluorometer, employing excitation at 486 nm and emission detection at 525 nm wavelengths [19].

Influence of zingerone on the nuclear condensation process in biofilm formation

To investigate nucleus condensation, a fluorescent dye designed for DNA-specific labeling known as 4',6-diamidino-2-phenylindole (DAPI) was employed. *C. albicans* and C1 isolate 1×10^7 cells/mL were allowed to adhere in 96 well plate at 37°C (90 min). The plates were rinsed with PBS. The cells underwent a treatment with MIC concentration of zingerone for biofilm formation. The cells without treatment were served as control. After the initial 24-hour incubation (37°C), the wells were subjected to another wash with sterile PBS. Subsequently, a treatment of DAPI (1 μ g/mL) was administered and kept in the dark at 30°C. Glass slides were then prepared to facilitate the analysis of nucleus staining, and observed under a confocal laser scanning microscope (Zeiss LSM 880) [21].

AO/EB (acridine orange/ethidium bromide) staining

Acridine orange/ethidium bromide (AO/EB) staining was utilized to distinguish between live and deceased cells of *C. albicans* ATCC 90028 and C1 isolate within the extensive cell population. This method involved scrutinizing the fluorescence pattern of cells, focusing on cell viability and the integrity of their membranes. Deceased cells allow ethidium bromide to enter, while viable cells permit the entry of acridine orange. 2×10^6 cells/mL were inoculated in the RPMI1640 medium containing MIC concentration of zingerone in 12 well plate. The plate underwent incubation at 35°C for 6 h. The cells underwent a washing procedure with sterile PBS three times. The AO/EB dual fluorescent staining solution was created by combining 100 μ g/mL of acridine orange (AO) and 100 μ g/mL of ethidium bromide (EB). Then, 10 μ l of the AO/EB staining solution was added and left to incubate for 15–20 minutes. The cells were examined using a fluorescent microscope (Nikon T1- SAM JAPAN) [23].

Membrane permeability assay

The membrane-impermeable red fluorescent dye, PI (Propidium Iodide), was frequently employed to assess damage to the plasma membrane to evaluate the impact on the membrane of *C. albicans* ATCC 90028 and C1 isolate, the cells were cultivated in the presence of zingerone. Post-treatment, the cells in their exponential phase were collected via centrifugation and subsequently washed thrice

with PBS to remove any cellular impurities. The cells were suspended in 50 μ l of PBS and then exposed to PI (100 μ g/mL) in darkness for at least 20–30 minutes. Afterward, the stained cells were observed under a fluorescence microscope at a magnification (Nikon T1- SAM JAPAN) [23].

Sorbitol assay

The MIC of zingerone on *C. albicans* ATCC 90028 and C1 isolate was determined utilizing the broth microdilution method in 96-well plates. To study effect of zingerone on the cell wall of *C. albicans* ATCC 90028 and C1 isolate osmotic support in the form of sorbitol (final concentration 0.8 M) was introduced into the culture medium. Aseptically sealed, following that, the plates were placed in an incubator at 35°C, and measurements were recorded at the intervals of 2 and 7 days. Notably, the observed higher MIC values in the sorbitol-supplemented medium compared with the standard medium suggested the potential role of the cell wall as a focal point for the tested product, given sorbitol's capacity to serve as a protective agent for fungal cell walls. The assay was conducted in triplicate [24].

Gene expression studies

Real-time PCR was used to evaluate quantification of signal transduction gene expression during the progression of serum-induced morphogenesis and the development of biofilms *C. albicans* ATCC 90028 and C1 isolate cells, with concentration 1×10^6 cells per/mL, were introduced and allowed to incubate for a duration of 90 minutes (37°C) in a medium 20 % serum supplemented by zingerone at 0.5 mg/mL. To investigate gene expression throughout the process of biofilm development in a 12-well plate, zingerone treatment at a concentration of 1 mg/mL was administered. Subsequently, the plates were placed at 37°C (24 h). The RNA extraction from planktonic cells was conducted by incubating *C. albicans* (1×10^7 cells/mL) with the MIC of zingerone, at 30°C for a duration of 4 h. Controls were established by using wells that did not receive any zingerone treatment. RNA extraction done using the RNeasy® Mini Kit from QIAGEN, USA. Subsequently, the extracted RNA was converted into complementary DNA (cDNA) using iScript® III Kit RT-PCR.

PCR was performed using the Uni SYBR GREEN Supermix® qPCR Kit. Master mix in 96-well PCR plates with initial denaturation for 3 minutes at 95°C, then 32 amplification cycles of primer annealing at 60°C for 20 seconds, primer extension at 72°C for 30 seconds (CFX 96 Real time System, Bio-Rad, USA) [15].

Primers

The primers used in the study were procured from Eurofins Genomics India Pvt. Ltd, Bangalore. For internal control, the housekeeping gene *ACT1N* was utilized (Table.1) [15].

Quantification of gene expression

The formula $2^{-\Delta\Delta CT}$ was applied to assess the transcript levels of selected genes. This experimental procedure was conducted in triplicate for robustness and reliability [15].

Silk worm survival assay

The silkworm, *Bombyx mori* early instar larvae were obtained from the Department of Zoology, Shivaji University, Kolhapur and reared on mulberry leaves. The fifth instar larvae were used for silkworm survival assay experiments. Fifth instar day one larvae weighing approximately 0.8 to 1 gm were taken in three sets and each set contained three larvae. In the treatment group, the larvae were subjected to a two-step injection process. Initially, they were injected

Table 1
Primers tailored for specific genes employed in quantitative real-time polymerase chain reaction (qPCR).

Primers	Sequence (5' → 3')
ACTIN-F	5'ATGGACGGTGAAGAAGTTGC 3'
ACTIN-R	5'ACCTCTTTTGGATTGGGCTTCA 3'
RAS1-F	5'GGCCATGAGAGAACAATATA 3'
RAS1-R	5'GTCTTTCCATTCTCTAATCAC 3'
PDE 2-F	5' ACCACCACCACTACTACTAC 3'
PDE 2-R	5' AAAATGAGTTGTTCCTGTCC 3'
BCY 1-F	5' CCC AAGCTTATGTCTAATCTCAACAGCA 3'
BCY 1-R	5' GGG CTGCAGTTAATGACCAGCAGTTGGGT 3'
EFG 1-F	5' TATGCCCCAGCAACAACACTG 3'
EFG 1-R	5' TTGTTGCTCTGCTGCTGTC3'
TEC 1-F	5' AGGTTCCCTGGTTTAAGTG 3'
TEC 1-R	5' ACTGGTATGTGTGGGTGAT 3'
ECE 1-F	5'CCCTCAACTTGCTCCTTCAACC3'
ECE 1-R	5'GATCACTTGTGGGATGTTGGTAA3'
CEK 1-F	5' AGCTATACAACGACCAATTA 3'
CEK 1-R	5' CATTAGCTGA ATGCATAGCT 3'
HST 7-F	5' ACTCCAACATCCAATATAACA 3'
HST 7-R	5' TTGATTGACGTTCAATGAAGA 3'
CPH1-F	5'ATGCAACACTATTTATACCTC 3'
CPH2-R	5'CGGATATTGTTGATGATGATA 3'
HWP1-F	5'TGGTGCTATTACTATTCCGG 3'
HWP1-R	5'CAATAATAGCAGCACCAGAAG 3'
MIG1-F	5'CTTCAACTAGCCTATATTCCGATGG 3'
MIG1-R	5'-CTTTCT GTAGGTACCAACAACACTAC 3'
NRG1-F	5'CACTCACTTGCAACCCC 3'
NRG1-R	5'GCCCTGGAGATGGTCTGA 3'
SOD1-F	5' TTGAACAAGAATCCGAATCC 3'
SOD1-R	5' AGCCAATGACACCACAAGCAG 3'
SOD2-F	5' ATGTTTTTCTATCAGATCATC 3'
SOD2-R	5' ACCACCACCTTGAGAGACAGGAGCC 3'
CAT1-F	5' CCCAGAAAGAGTTGTCCACGC 5'
CAT1-R	5' CCATGATGGGTAGTTACCAGCAGC 5'
MCA1-F	5' TGGTACTGCCACTGGTGCTA 3'
MCA1-R	5' TGGGAAGCAGATAATTGTGG 3'

with *C. albicans* and C1 isolate, followed by the injection of zingerone compound (test). In the positive control group, larvae received injections of *C. albicans* cells and C1 isolate alone, while in negative control, they were injected with sterile distilled water only. Also, one group was injected with 2 % DMSO (vehicle control). All treatments (25 μ l) were given to silkworm larvae with the help of a 27-gauge needle syringe through 3rd abdominal leg and then silkworms were maintained for survival period observation at laboratory conditions at 24° C with 70 % relative humidity (Table 2) [25].

Statistical analysis

The values provided represent the means and standard deviations derived from three distinct observations. Statistical analysis was conducted using the student's t-test in order to compare the values between the control and treatment groups. A significance level of P < 0.05 was taken into account as Significant from a statistical standpoint [14]. *In vivo* experimental data was analyzed by using Graphpad prism 6.0 software with help of Kaplan meier curve.

Table 2
Specification of groups used in the experiment.

Group no.	Type of administration	Type of group
Group-1	Larvae injected with sterile D/W	Negative control
Group -2	Larvae injected with 1×10^6 cells/mL of <i>C. albicans</i>	Positive control
Group -3	Larvae injected with <i>C. albicans</i> and subsequently with zingerone	Test
Group -4	Larvae injected with DMSO	Vehicle control

Table 3
Zingerone influence growth, adhesion, morphogenesis, and biofilm formation in *C. albicans* ATCC 90028 and C1 isolate.

Molecule	MIC (mg/mL)					
	Planktonic growth	Developing biofilm	Mature biofilm	Adhesion	Yeast-to-hyphal morphogenesis	MFC
<i>C. albicans</i> ATCC 90028	2	2	NA	0.125	1	4
<i>C. albicans</i> C1 isolate	4	4	NA	0.25	2	NA

NA not achieved up to 4 mg/mL; MFC, minimum fungicidal concentration; MIC, minimum inhibitory concentration.

Results

Zingerone inhibited the planktonic growth of *C. albicans*

Zingerone exhibited inhibitory activity against the planktonic growth of *C. albicans* ATCC 90028 and C1 isolate in a concentration-dependent manner. The MIC, which represents the Minimum Inhibitory Concentration at which 50 % growth inhibition, it was observed that at 2 mg/mL for *C. albicans* ATCC 90028 and 4 mg/mL for the C1 isolate zingerone reduced the 50 % growth, compared to the control (Fig. 2a; Table 3). The Minimum Fungicidal Concentration (MFC) for *C. albicans* ATCC 90028 was determined to be 4 mg/mL, while no MFC was observed for the C1 isolate at the concentrations tested. Additionally, exposure to zingerone resulted in a significant reduction in cell viability, with a 99% decrease in *C. albicans* ATCC 90028 viability (Fig. 2b; Table 3) Amphotericin B inhibited the planktonic growth of *C. albicans* ATCC 90028 at a concentration of 0.39 µg/mL, whereas the C1 isolate was inhibited at a concentration of 0.78 µg/mL. The MFC for *C. albicans* ATCC 90028 was achieved at 0.7 µg/ml and for C1 isolate it was 1.5 µg/ml (Table 4, Fig. S1, S2).

Zingerone eradicated *C. albicans* cells

The fungicidal activity of zingerone is clearly illustrated in Fig. 2b. Based on the kill curve experimental results, treatment with zingerone at a concentration of 4 mg/mL led to the complete eradication of *C. albicans* ATCC 90028 cells, achieving a 100% kill rate within 480 minutes. However, zingerone did not exhibit a similar killing effect on the C1 isolate, as growth was still observed on the plates even after 480 minutes of treatment. This suggests that while zingerone is effective against the reference strain, its effect on the C1 isolate differs (Fig. 2c).

Zingerone inhibited metabolic activity of *C. albicans*

The addition of zingerone significantly affected the metabolic of *C. albicans* ATCC 90028 and the C1 isolate, with the most pronounced effects observed at concentrations up to 0.125 mg/mL. Analysis using the XTT-metabolic assay revealed a 50% reduction in metabolic

activity at this concentration. Based on the reduced metabolic activity, it was concluded that zingerone effectively impaired the metabolic activity of *C. albicans*. In contrast, the C1 isolate required a higher concentration of 0.25 mg/mL to observe similar inhibition of metabolic activity. At higher zingerone concentrations, specifically 2 mg/mL and 4 mg/mL, metabolic activity was reduced by up to 90 % and 97 % respectively (Fig. 2d; Table 3). Amphotericin B inhibited the metabolic activity of *C. albicans* ATCC 90028 at a concentration of 13 µg/mL, and at 25 µg/mL it inhibited the metabolic activity of the C1 isolate (Table 4, Fig. S1).

Zingerone inhibited of yeast- to- hyphal morphogenesis in *C. albicans*

Zingerone exhibited a concentration-dependent inhibition of the yeast-to-hyphal transition in *C. albicans* induced by fetal bovine serum (FBS). Treatment with zingerone significantly inhibited this critical morphological transformation. Specifically, at a concentration of 1 mg/mL, 50% inhibition of yeast-to-hyphal morphogenesis was observed in *C. albicans* ATCC 90028 cells, while the C1 isolate required 2 mg/mL concentration for 50 % inhibition of yeast-to-hyphal morphogenesis. At a higher concentration of 4 mg/mL, complete inhibition of this morphological transition was achieved in both strains, demonstrating strong ability of zingerone to prevent hyphal development, which is crucial for pathogenicity of *C. albicans* (Fig. 3a, 3b Table 3). The MIC of amphotericin B for yeast- to- hyphal morphogenesis was observed at 0.7 µg/mL for *C. albicans* 90028 and 1.5 µg/mL for C1 isolate (Table 4, Fig. S1, S2).

Zingerone inhibited biofilm formation observed by confocal microscopy

The qualitative assessment of biofilm formation in *C. albicans* ATCC 90028 and the C1 isolate, using acridine orange dye, revealed dense biofilm structures in the untreated (control) cells. However, treatment with zingerone resulted in a marked reduction in biofilm thickness, which appeared to be concentration-dependent. This reduction indicates that zingerone effectively disrupts biofilm formation, reducing its overall density. To further investigate and visualize these effects, confocal laser scanning microscopy (CLSM) was utilized. The CLSM images, shown in Fig. 3c, clearly demonstrate the structural changes and decreased biofilm density following zingerone treatment, highlighting its potential as a biofilm inhibitor in *C. albicans*.

Zingerone inhibited biofilm formation

Biofilm quantification using the crystal violet assay revealed that at concentrations of 2 mg/mL and 4 mg/mL, zingerone caused a 50% reduction in biofilm formation by *C. albicans* ATCC 90028 and the C1 isolate compared to the untreated control biofilms (Fig. 3d). Microscopic observation of 48-hour biofilms grown on polystyrene surfaces and stained with 0.4% crystal violet showed a significant reduction in biofilm cells in the zingerone-treated groups (Fig. 4a), indicating a strong inhibitory effect on biofilm development.

Table 4
The antifungal efficacy of amphotericin B against *C. albicans* ATCC 90028 and *C. albicans* C1 isolate showing the minimum inhibitory concentration for planktonic growth, biofilm Formation, mature biofilm, adhesion, morphogenesis, and minimum fungicidal concentrations for Planktonic Growth.

Molecule	MIC (µg/mL)				
	Planktonic growth	Developing biofilm	Mature biofilm	Adhesion	MFC
<i>C. albicans</i> ATCC 90028	0.39	0.39	0.78	13	0.78
<i>C. albicans</i> C1 isolate	0.78	0.78	1.5	25	1.5

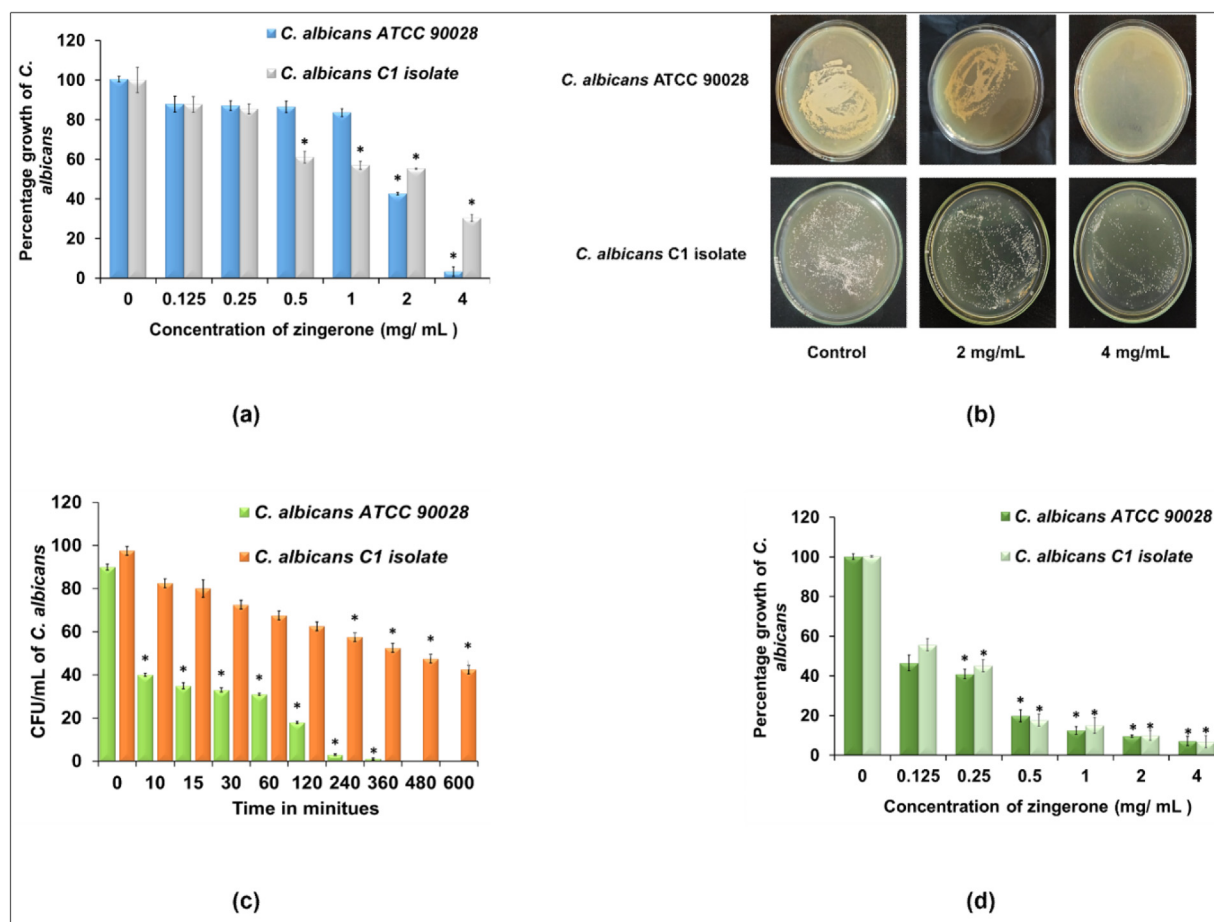


Fig. 2. (a) Effect of zingerone against planktonic growth of *C. albicans* ATCC 90028 and C1 isolate. The experiment was performed in triplicates. * to represent the statistical significance which means p-value is less than 0.05. (b) Minimum Fungicidal Concentration (MFC) of zingerone against *C. albicans* ATCC 90028 and C1 isolate cells (c) Time-kill kinetics assay depicting zingerone was added to media containing a *C. albicans* ATCC 90028 and C1 isolate. The log CFU/mL for zingerone was determined at time 0 and at subsequent time points up to 24 hours (d) Effect of zingerone on adhesion of *C. albicans* ATCC 90028 and C1 isolate. The experiment was performed in triplicates. * to represent the statistical significance which means p-value is less than 0.05.

Scanning electron microscopy (SEM) images further confirmed this inhibition. In the control samples, a complex three-dimensional biofilm structure with long hyphae forming a multilayered network was visible. In contrast, the zingerone-treated samples showed no significant hyphal formation, highlighting effectiveness of zingerone in completely disrupting biofilm development after 24 hours of treatment in both the strains (Fig. 4b). Amphotericin B inhibited biofilm development at 0.39 $\mu\text{g/mL}$ in *C. albicans* ATCC90028 and in C1 isolate it inhibited biofilm at 0.78 $\mu\text{g/mL}$. It was also observed that zingerone did not inhibited mature biofilm of *C. albicans* ATCC90028 and C1 isolate (Table 4). On the other hand, Amphotericin B inhibited mature biofilm of *C. albicans* ATCC90028 at 0.7 $\mu\text{g/mL}$ and the C1 isolate at 1.5 $\mu\text{g/mL}$ (Table 4, Fig. S1, S2).

Effect of zingerone on membrane permeability

Propidium iodide (PI) is a fluorescent intercalating dye that binds to DNA but is unable to penetrate healthy cells. However, when the cell membrane is compromised, PI enters the cytoplasm and binds to DNA, producing bright red fluorescence under a fluorescence microscope. Fluorescence microscopy revealed an increase in the number of damaged *C. albicans* ATCC 90028 and C1 isolate cells as the concentration of zingerone increased from 1 mg/mL to 4 mg/mL. The PI uptake assay further confirmed a significant rise in PI-positive cells following zingerone treatment compared to the control, indicating

that zingerone induces cell membrane damage in *C. albicans* ATCC 90028 and the C1 isolate (Fig. 4c).

Zingerone inhibited ergosterol biosynthesis in *C. albicans*

Ergosterol is a critical lipid component of the membrane in *C. albicans* ATCC 90028 and the C1 isolate, playing a vital role in maintaining membrane fluidity, permeability, and overall integrity. As a key structural element, ergosterol is essential for the proper function of fungal membranes, making it an important target for understanding fungal biology and developing antifungal strategies. The results of this study demonstrate that zingerone treatment had a significant, dose-dependent impact on the ergosterol profile of both strains.

In *C. albicans* ATCC 90028, treatment with zingerone at concentrations of 2 mg/mL and 4 mg/mL resulted in a complete inhibition of ergosterol content, indicating a severe disruption of ergosterol synthesis or maintenance. Similarly, a concentration-dependent inhibition of ergosterol was observed in the C1 isolate, though the degree of inhibition varied across different concentrations. These findings suggest that zingerone interferes with ergosterol at both high and low concentrations, with higher concentrations leading to more pronounced disruptions.

Overall, the results highlight zingerone's ability to impair the ergosterol biosynthetic pathway, disrupting fungal membrane integrity in a dose-dependent manner. The broader range of ergosterol reduction, even at lower zingerone concentrations, underscores its

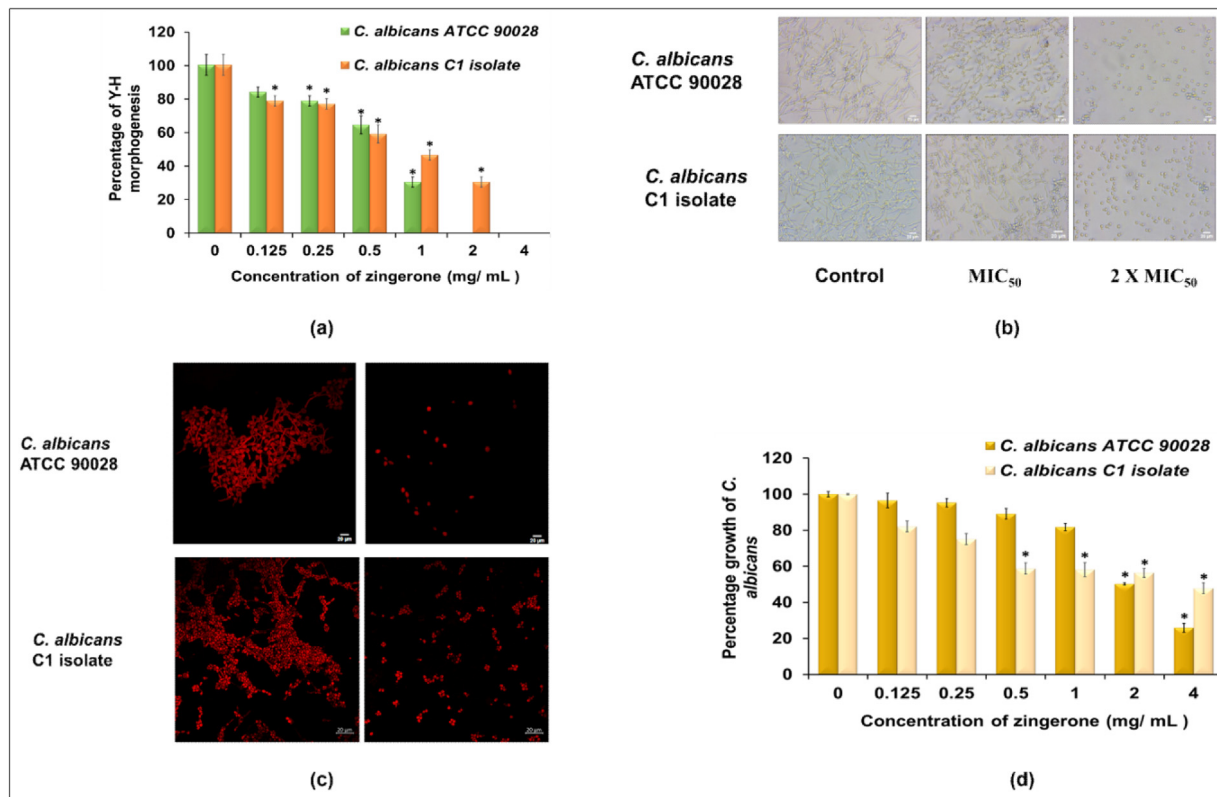


Fig. 3. (a) Inhibition of Yeast-to-hyphal morphogenesis in *C. albicans* ATCC 90028 and C1 isolate by zingerone. (b) Microscopic observation of inhibition of Yeast-to-hyphal morphogenesis in *C. albicans* ATCC 90028 and C1 isolate after the treatment of zingerone. (c) Inhibition of developing biofilm visualized by Confocal microscopy in *C. albicans* ATCC 90028 and C1 isolate after the treatment of zingerone (d) Effect of zingerone on developing biofilm of *C. albicans* ATCC 90028 and C1 isolate. Biofilm was quantified by crystal violet assay. The experiment was performed in triplicates. Error bars indicate standard deviations. * to represent the statistical significance which means p-value is less than 0.05.

potential as a potent antifungal agent in both standard and clinical isolates of *C. albicans*. (Fig. 4d, 5a).

The effect of zingerone on the cell wall of *C. albicans*

The focus of the sorbitol assay was to ascertain the minimum inhibitory concentration of zingerone in the presence and absence of 0.8 M sorbitol, which served as an osmotic protector. The results obtained suggested that the MIC of zingerone against *C. albicans* ATCC 90028 doubled from 2 mg/mL to 4 mg/mL in the presence of sorbitol. While in case of C1 isolate it was increased from 4 mg/mL to 8 mg/mL. This substantial increase in the MIC value suggests that sorbitol, acting as an osmotic protector, influenced the effectiveness of zingerone against *C. albicans* ATCC 90028 and C1 isolate (Table 5).

Effect of zingerone on nuclear condensation in *C. albicans*

The study focused on examining the impact of zingerone on genetic material integrity in *C. albicans* ATCC 90028 and C1 isolate. The key points including how zingerone affects the integrity of DNA in *C. albicans* in both the strains. Cells with damaged DNA may face challenges in properly repairing their DNA, which can ultimately result in cell death. To assess DNA integrity, DAPI staining was employed. DAPI binds to the A-T rich regions of DNA, enabling the visualization of nuclear condensation and fragmentation [26]. Zingerone-treated *C. albicans* ATCC 90028 and C1 isolate cells exhibited nuclear abnormalities at MIC concentration. These observed abnormalities may be attributed to factors such as oxidative stress and apoptosis, suggesting that zingerone treatment had a notable impact on DNA integrity and cellular health compared to untreated cells. This succinctly conveys the main findings and implications of the study

regarding the effects of zingerone on *C. albicans* DNA integrity (Fig. 5b).

Zingerone inhibited cell division cycle of *C. albicans*

Flow cytometry analysis revealed that zingerone at MIC significantly affected the cell cycle of *C. albicans* ATCC 90028 and the C1 isolate, causing cell cycle arrest at specific stages. In the control group of *C. albicans* ATCC 90028, 88 % of the cells were found in the G0/G1 phase, 10 % in the S phase, and 2 % in the G2/M phase. Similarly, in the control group of the C1 isolate, 52 % of cells were in the G0/G1 phase, 37.3 % in the S phase, and 12.5 % in the G2/M phase, representing the normal cell cycle distribution. Upon treatment with zingerone, a notable shift occurred, with 51 % of *C. albicans* ATCC 90028 cells and 58.7 % of C1 isolate cells being arrested in the G0/G1 phase, indicating a significant accumulation of cells at this stage. In contrast, the S phase showed minimal disruption, suggesting that zingerone had little impact on this phase. Furthermore, zingerone induced apoptosis in *C. albicans* ATCC 90028 cells, with approximately 20 % of the cell population undergoing programmed cell death following treatment (Fig. 5c, 5d).

Zingerone induced the production of intracellular reactive oxygen species in *C. albicans*

Intracellular reactive oxygen species (ROS) levels were assessed using the H₂DCF-DA probe, which is non-fluorescent until oxidized by ROS to form the highly fluorescent DCF, emitting green fluorescence. In this study, treatment with zingerone led to a marked increase in the number of green fluorescent cells in both *C. albicans* ATCC 90028 and the C1 clinical isolate. This increase in fluorescence

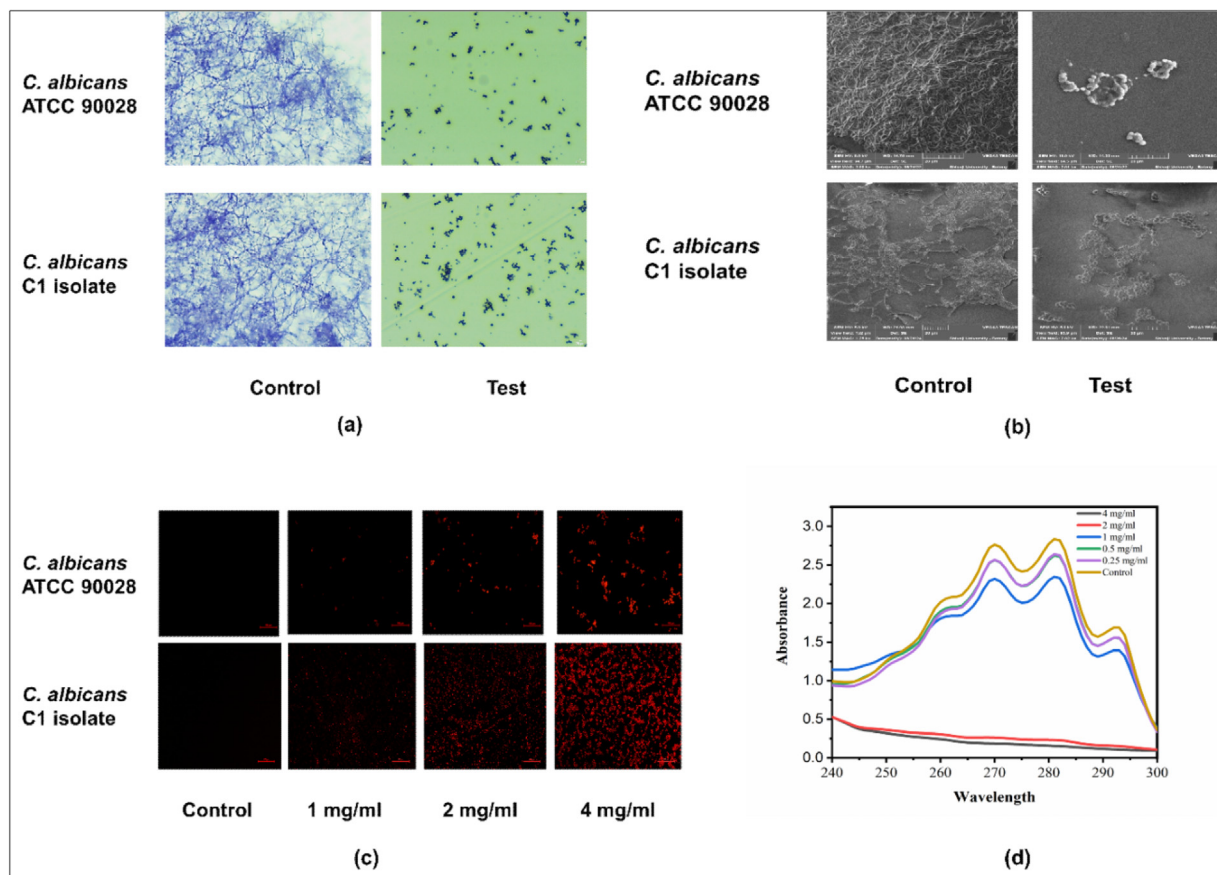


Fig. 4. (a) Inverted light microscopic images of *C. albicans* ATCC 90028 and C1 isolate, biofilms stained with 0.4 % crystal violet showing maximum inhibition in test treated groups compared to untreated control (Scale bar: 20 μ m). (b) Scanning electron micrographs after 24 h incubation on silicon Foley's urinary catheter piece. Biofilm without treatment and inhibition of *C. albicans* ATCC 90028 and C1 isolate biofilm development in presence of zingerone. (c) PI staining of *C. albicans* ATCC 90028 and C1 isolate cells treated with different concentrations of zingerone visualized by fluorescent microscopy (Scale bar: 100 μ m) (d) Effect of zingerone on ergosterol biosynthesis of *C. albicans* ATCC 90028. Ergosterol was extracted from cells treated by 0.25 to 4 mg/mL concentrations of zingerone and measured by the spectrophotometric method.

corresponds to elevated ROS levels within the cells, indicating that zingerone induces intracellular oxidative stress. In the untreated control group (negative control), minimal fluorescence was observed, which suggests that baseline ROS levels were low under normal conditions without external stress. In contrast, cells treated with zingerone exhibited significantly higher fluorescence, implying that the compound promoted ROS accumulation, leading to oxidative stress. The positive control group, subjected to known ROS inducers, displayed the highest intensity of green fluorescence, signifying severe oxidative stress and maximal ROS generation. This elevated fluorescence served as a reference point to demonstrate the extent of oxidative stress in comparison to other groups.

These findings indicate that while zingerone induces ROS production and triggers an oxidative stress response, the level of ROS generated is lower than in the positive control. This suggests that zingerone-mediated ROS accumulation may contribute to its antifungal activity by damaging cellular structures in a controlled manner, without reaching the extreme levels of oxidative stress seen in the positive control. The results therefore reveal an additional mechanism by which zingerone exerts its antifungal effects, complementing its previously established roles in inhibiting biofilm formation and modulating gene expression (Fig. 6a).

Zingerone modulated gene expression involved in morphogenesis

To investigate the effect of zingerone on the inhibition of the morphogenic switch at the gene expression level, qRT-PCR was carried out. The relative fold changes, shown in Fig. 6b, reflect the altered

gene expression levels in *C. albicans* cells treated with zingerone, with expression normalized to the housekeeping actin gene. Following zingerone treatment, the negative regulators of hyphal development, *NRG1* and *MIG1*, showed a differential expression pattern between the two strains. In *C. albicans* ATCC 90028, *NRG1* and *MIG1* were downregulated by 1.07-fold and 1.2-fold, respectively. However, in the C1 clinical isolate, these genes were significantly upregulated, with a 4.3-fold and 10.84-fold increase, respectively. Additionally, another key negative regulator, *TUP1*, exhibited upregulation in both strains, with a 2.19-fold increase in ATCC 90028 and a remarkable 28.7-fold increase in the C1 isolate. The expression of *BCY1*, a gene involved in morphogenesis, was downregulated in *C. albicans* ATCC 90028 following zingerone treatment, while it was upregulated in the C1 isolate. Moreover, the *CEK1* gene was upregulated in both strains upon zingerone treatment compared to the non-treated control.

Zingerone altered gene expression during biofilm formation

To explore the inhibitory effects of zingerone on biofilm formation in *C. albicans* ATCC 90028 and the C1 clinical isolate, quantitative real-time PCR was performed to assess the expression of key genes associated with the Ras1-cAMP-Efg1 and Cek1-MAPK pathways. The *RAS1* gene, which plays a crucial role in *C. albicans* pathogenicity, showed significant downregulation following zingerone treatment, with expression reduced by 6.27-fold in *C. albicans* ATCC 90028 and 5.89-fold in the C1 isolate.

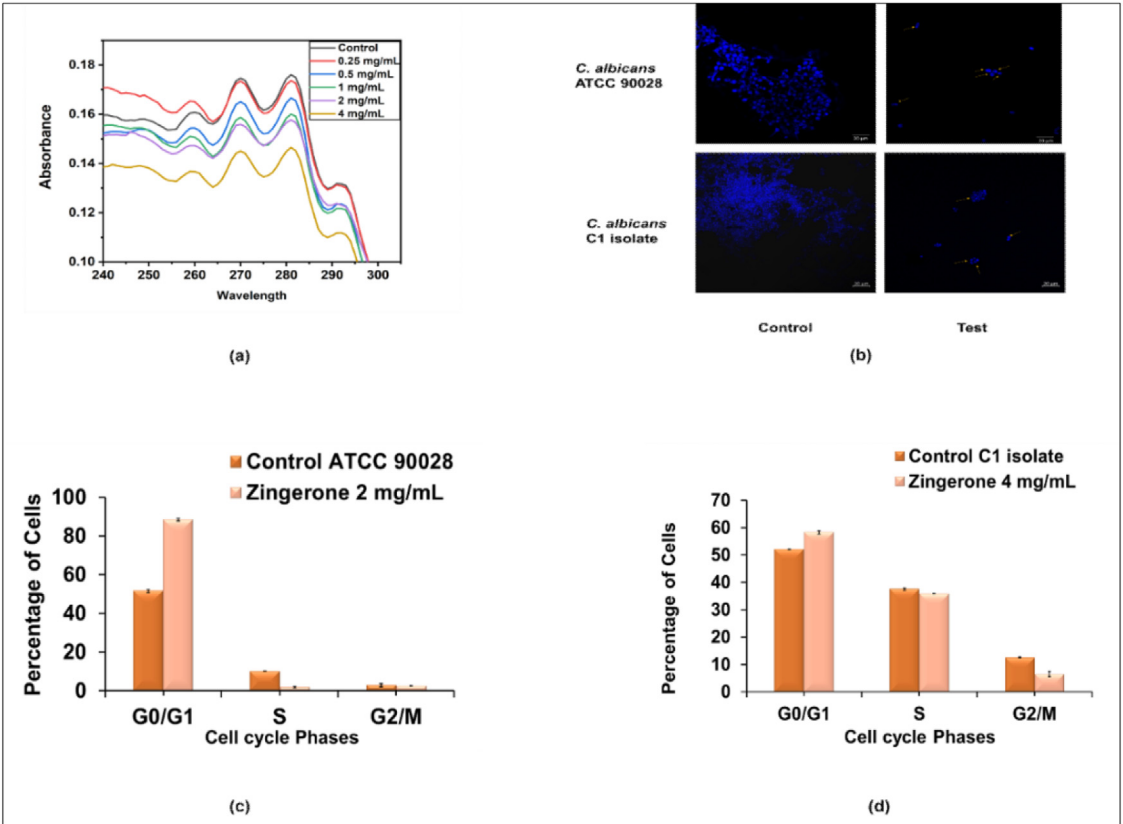


Fig. 5. (a) Effect of zingerone on ergosterol biosynthesis of C1 isolate. Ergosterol was extracted from cells treated by 0.25 to 4 mg/mL concentrations of zingerone and measured by the spectrophotometric method. (b) Nuclear condensation in *C. albicans* ATCC 90028 and C1 isolate cells treated with MIC concentration of zingerone and without zingerone was detected by DAPI staining. Nuclear condensation was visualized with confocal microscopy (scale bar: 20 μm). (c) Flow cytometric analysis of cell cycle arrest after the treatment of zingerone with propidium iodide staining. The bars indicate the percent of cells in G1, S, and G2/M phases in control cells and in MIC test i.e. zingerone treated cells. zingerone arrest *C. albicans* ATCC 90028 cells at G0/G1- phase. (d) Flow cytometric analysis of cell cycle arrest after the treatment of zingerone with propidium iodide staining. The bars indicate the percent of cells in G1, S, and G2/M phases in control cells and in MIC test i.e. zingerone treated cells. zingerone arrest C1 isolate cells at G0/G1- phase.

The *PDE2* gene, essential for maintaining hyphal growth and cell wall integrity, displayed differential regulation it was down-regulated by 2.40-fold in ATCC 90028 and by 10.81-fold in the C1 isolate. Similarly, genes involved in the cAMP pathway, including *EFG1* and *TEC1*, were consistently downregulated in both strains. Specifically, *EFG1* was reduced by 12.47 -fold and 49.84-fold, and *TEC1* by 1.33-fold and 7.74-fold in *C. albicans* ATCC 90028 and C1, respectively. On the other hand, *BCY1* was downregulated in case of *C. albicans* ATCC 90028 by 1.17- fold and in C1 isolate it was upregulated by 4.49- fold.

Further downregulation was observed in genes downstream of *EFG1*, such as *ECE1* and *HWP1*, which are critical for hyphal formation and biofilm development. *ECE1* was downregulated by 2.76 -fold in ATCC 90028 and 13.68-fold in C1, while *HWP1* showed a 1.54- fold and 3.43-fold reduction in both strains respectively.

In the Cek1-MAPK pathway, the *CPH1* and *HST7* genes, which regulate hyphal development, were also downregulated. In ATCC 90028, *CPH1* decreased by 8.56-fold and *HST7* by 11.26-fold, while in the C1 isolate, they were downregulated by 21-fold and 35-fold, respectively. The *CEK1* gene experienced a substantial reduction

in expression, downregulated by 21.42-fold in ATCC 90028 and 2.48 -fold in C1.

Additionally, the hyphal suppressor genes *NRG1* and *MIG1* were affected. *NRG1* and *MIG1* were downregulated by 2.11-fold and 2.29-fold in *C. albicans* ATCC 90028. In case of C1 isolate *NRG1* was down-regulated by 8.05- fold while *MIG1* was upregulated by 1.55- fold (Fig. 6c).

Zingerone altered the gene expression involved in ROS production

The study investigated the expression of genes involved in oxidative stress protection, including *SOD1*, *SOD2*, *CAP1*, *CAT1*, and *MCA1*, to validate the observed ROS production and its impact on the cell cycle. After zingerone treatment, both *SOD2* and *CAT1* showing 1.18 -fold and 1.67 -fold reduction in *C. albicans* ATCC 90028, on the other hand a *SOD2* was downregulated by 1.30 -fold and *CAT1* was upregulated by 1.67 -fold in the C1 isolate, respectively. Notably, *SOD1* also exhibited significant overexpression in C1 isolate post-treatment. Interestingly, *MCA1*, a gene linked to apoptosis, demonstrated a 1.38-fold upregulation in *C. albicans* ATCC 90028 and downregulated by a 1.54-fold in the C1 isolate following zingerone exposure. These distinct expression patterns highlight zingerone had varying effects on genes associated with oxidative stress protection and apoptosis. The observed changes in gene expression likely contribute to the differences in ROS generation and cell cycle dynamics, offering deeper insight into the molecular mechanisms through which zingerone impacts *C. albicans* (Fig. 6d).

Table 5
MIC values of zingerone in the absence and presence of sorbitol (0.8 M).

Name of organism	MIC in presence of sorbitol	MIC in absence of sorbitol
<i>C. albicans</i> ATCC 90028	4 mg/mL	2 mg/mL
<i>C. albicans</i> C1 isolate	8 mg/mL	4 mg/mL

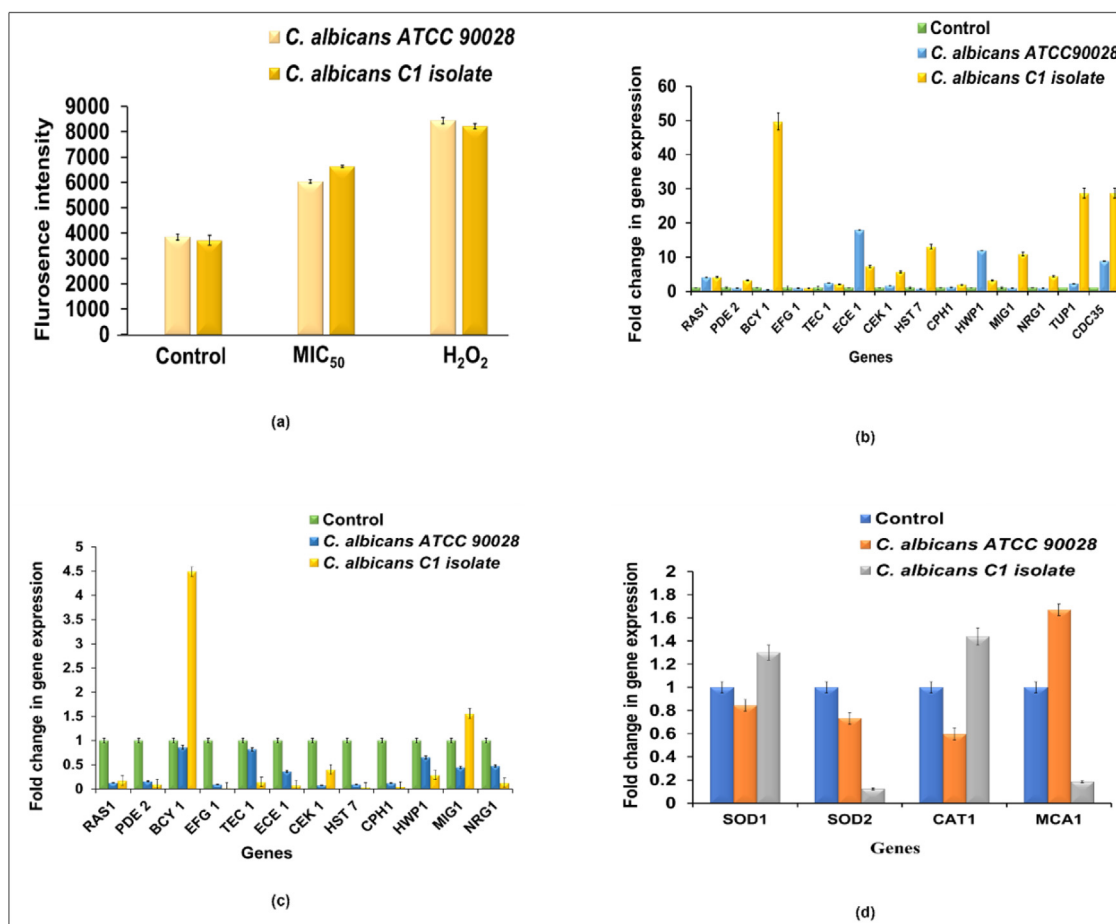


Fig. 6. (a) Effect of zingerone on endogenous ROS accumulation of *C. albicans* ATCC 90028 and C1 isolate. (b) Gene expression profile after the treatment of zingerone during *C. albicans* ATCC 90028 and C1 isolate yeast- to- hyphal morphogenesis. (c) Gene expression profile after the treatment of zingerone during *C. albicans* ATCC 90028 and C1 isolate biofilm formation. (d) gene expression profile involved in preventing ROS damage in *C. albicans* ATCC 90028 and C1 isolate.

Detection of cell viability after the treatment of zingerone

AO and EB serve as stains with distinct affinities AO penetrates live cells, emitting a green fluorescence, while EB enters deceased cells, exhibiting a red fluorescence. In evaluating the viability of *C. albicans* ATCC 90028 and C1 isolate post-exposure to zingerone, we employed the AO/EB dual staining method. In the control group, a significant density of live cells, indicated by their green coloration, was observed. However, after treatment with the of zingerone, the cells displayed red fluorescence, indicating their demise. This shift from green to red fluorescence following zingerone treatment suggests a notable impact on cell viability. The change in staining patterns, from live to deceased cell indicators, emphasizes the effectiveness of zingerone in inducing cell death within the *C. albicans* ATCC 90028 population on the other hand zingerone did not have killing effect on C1 isolate hence a significant density of live cells, indicated by their green coloration, was observed (Fig. 7a).

Zingerone did not cause significant hemolysis

In the context of assessing potential drug-induced immune hemolytic anemia, an *in vitro* hemolysis study was conducted as a precaution before considering any pharmaceutical applications. This study adhered to the guidelines set forth by the European Medicines Agency [27]. Hemolysis is commonly employed as the initial assessment of toxicity in drug development because it can be indicative of cytotoxicity. Cell membrane disruption is often a primary cause of

toxicity. In the case of zingerone, the study revealed that it induced 2% hemolysis of human red blood cells (RBCs) when applied at a concentration of 4 mg/mL. However, at lower concentrations tested in the series, zingerone did not cause hemolysis. This information is essential for evaluating the safety profile of zingerone and its potential for pharmacological applications while considering the risk of immune hemolytic anemia as a rare side effect (Fig. 7b).

Zingerone increased the survival rate of silk worms

The *in vivo* antifungal efficacy was evaluated using a silkworm model. This model is commonly employed in scientific research to assess the effectiveness of antifungal treatments in a living organism, providing valuable insights into the potential therapeutic benefits of the tested substances. The results showed that silkworms treated with a combination of *C. albicans* and zingerone at its MIC concentration were able to survive for up to 72 hours. This suggests that the use of zingerone alongside *C. albicans* ATCC 90028 and C1 isolate had a protective effect on the survival of the silkworms. In this experiment, prolonging their lifespan compared to those treated with *C. albicans* alone. In contrast, the silkworm larvae that were treated with only *C. albicans* ATCC 90028 and C1 isolate did not survive beyond 24 hours of treatment, indicating that the fungal infection caused mortality within this time frame. On the other hand, the silkworm larvae treated with distilled water (negative control) not only survived but also progressed to the pupation stage within their cocoons. This suggests that the negative control group did not

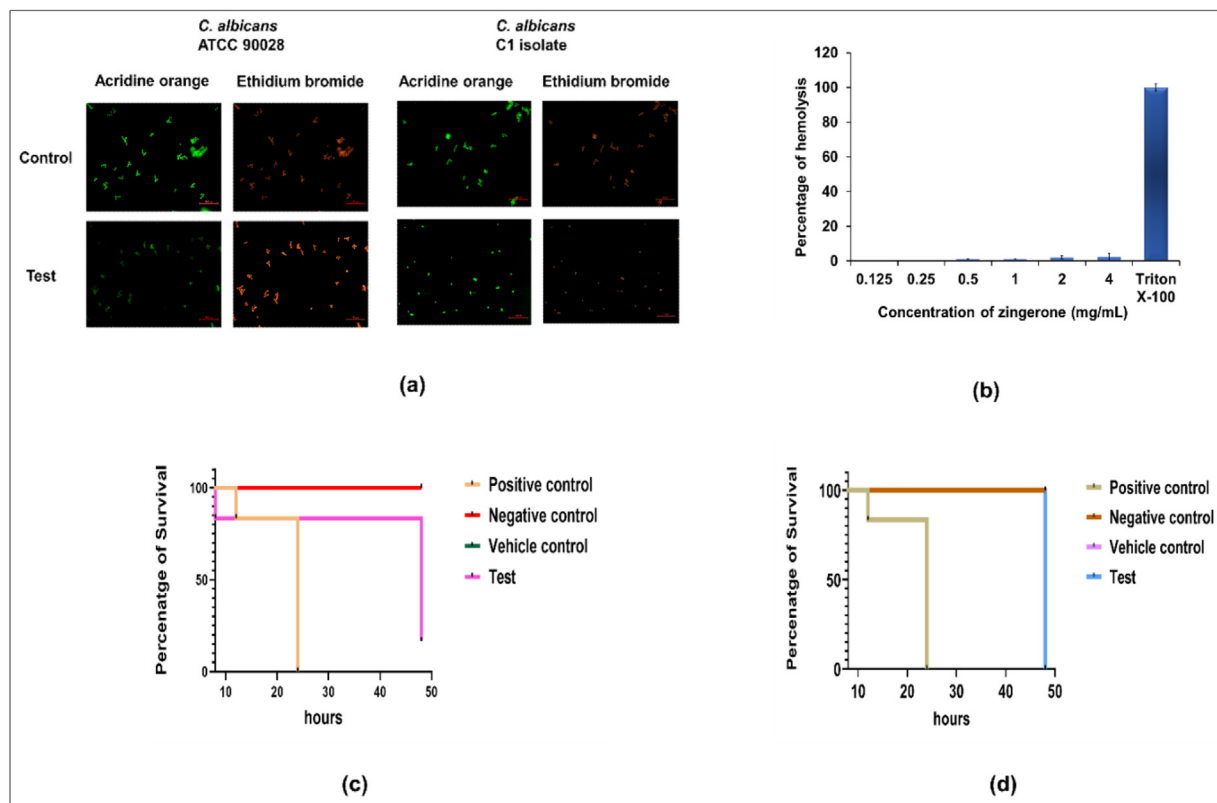


Fig. 7. (a) AO and EB staining showing live cells, emitting a green fluorescence and dead cells, exhibiting a red fluorescence with and without treatment of zingerone. (b) Hemolytic activity helps to analyse the effect of zingerone on human Red Blood Cells (RBC). Zingerone exhibited no significant hemolysis up to 4 mg/mL concentration. (c) Kaplan-Meier curve survival curves silkworms infected with *C. albicans* ATCC 90028 and treated with zingerone showed enhanced survival period compared to infected silkworm. (d) Kaplan-Meier curve survival curves silkworms infected with C1 isolate and treated with zingerone showed enhanced survival period compared to infected silkworm.

experience significant adverse effects and continued with their normal developmental process. Additionally, silkworms that were injected with a 2% DMSO (Dimethyl Sulfoxide) solution also exhibited extended survival, lasting up to 72 hours. This indicates that the DMSO treatment did not result in substantial harm to the silkworms, and they were able to survive for the duration of the experiment, similar to the group treated with *C. albicans* and zingerone (Fig. 7c, 7d).

Discussion

C. albicans has emerged as a significant threat to immunocompromised individuals, such as those with HIV/AIDS, cancer, or prolonged exposure to high doses of antibiotics [28]. *C. albicans* can form biofilms on both living and non-living surfaces, which significantly contribute to its increased resistance to antifungal medications. This highlights the urgent need for discovering novel antifungal agents. Natural compounds, particularly those derived from herbs and medicinal plants, are gaining recognition as promising candidates for developing new antifungal therapies [29]. Medicinal plants and phytochemicals are often highlighted as innovative antifungal treatments for effectively managing *Candida* infections, with numerous reports indicating they typically cause minimal or no adverse effects [30]. Zingerone possesses an extensive array of bioactive properties like antioxidant, anti-inflammatory, anticancer and antimicrobial activity. Previously the antifungal activity of zingerone was tested against *C. albicans* and to ascertain anti-fungal effectiveness for a particular application, more research is required [12,31].

In current study, we comprehensively investigated the antifungal effects of zingerone on *C. albicans* ATCC 90028 and the C1 isolate using both *in vitro* and *in vivo* approaches. The preliminary study

revealed that planktonic growth was inhibited at a concentration of 4 mg/mL in both strains (Fig. 2a Table 3). A major pathogenic trait of *C. albicans* is its ability to form biofilms, which not only enhance adhesion to body surfaces but also act as a protective shield against environmental stress and the host immune system, often resulting in increased tolerance to antifungal agents [32]. Our research findings indicate that zingerone effectively inhibits the yeast-to-hyphae (Y-H) transition and prevents biofilm formation. Zingerone demonstrated significant biofilm inhibitory activity against *C. albicans* ATCC 90028 at a concentration of 2 mg/mL and against the C1 isolate at 4 mg/mL. The inhibitory effects of zingerone on adhesion and biofilm development, as observed under scanning electron microscopy and confocal microscopy, provide strong evidence of its anti-*C. albicans* properties (Fig. 3, 4). In the scanning electron microscopy (SEM) analysis, the control sample exhibited a complex three-dimensional structure with elongated hyphae forming a multilayered network. In contrast, the treated sample showed a marked reduction in significant hyphal formation in both *C. albicans* ATCC 90028 and the C1 (Fig. 4b). This membrane serves as a physical barrier and plays key roles in cell wall synthesis, nutrient transport, influencing the virulence behavior, and structural transitions of the organism [33]. It was found that at higher concentration tested, there was a significant accumulation of PI, indicating a maximum effect on membrane integrity. However, as the concentrations of zingerone increased more pronounced membrane damage and lesions following PI staining were observed in both the tested strains (Fig. 4c). Ergosterol serves a role in fungi that is somewhat analogous to the role of cholesterol in mammalian cells. Ergosterol is a crucial element of fungal cellular membranes and it plays several essential functions in maintaining membrane structure and function like it maintains permeability, fluidity, and integrity of the membrane in fungi [26]. Zingerone inhibited ergosterol synthesis in a

dose-dependent manner in both the strains. In *C. albicans* ATCC 90028 at higher concentrations complete inhibition of ergosterol synthesis was observed. While in case of C1 isolate there was no complete inhibition was observed (Fig. 4d, 5a). To explore zingerone's impact on the cell wall, we conducted an assay using sorbitol, prevalent osmotic shielding agent utilized for the stabilization of fungal protoplasts. The increase in the MIC value observed in sorbitol-containing medium compared to MIC value in standard medium in both the strain (Table 5). Cells shielded with sorbitol can thrive even in the presence of inhibitors targeting fungal cell wall synthesis [24], whereas their growth would typically be hindered without sorbitol. Hence zingerone may have ability to inhibit the cell wall biosynthesis. The minor groove of DNA contains A-T rich regions. DAPI binds to the DNA at A-T rich region [34]. Binding of zingerone to the DNA may cause abnormalities in cell division which ultimately results in apoptosis. Hence it was observed that enhanced DAPI staining after the treatment of zingerone due to DNA damage. Zingerone binds with DNA, resulting in nuclear condensation and DNA fragmentation in *C. albicans* ATCC 90028 and C1 isolate (Fig. 5b). All multicellular organisms use the evolutionary conserved process called apoptosis to control their development and differentiation [35]. Key features of apoptosis include cellular shrinkage, nuclear fragmentation, chromatin condensation, and generation of apoptotic bodies; these are ultimately engulfed by neighboring cells or macrophages, causing no inflammatory reaction. Apoptosis is tightly controlled by a network of genes and signaling pathways to ensure its proper execution. Zingerone induced apoptosis in *C. albicans* cells. Cell cycle and apoptosis are interconnected processes and play essential roles in cell survival, growth, and reproduction. Zingerone arrested *C. albicans* at G1 phase of cell cycle may be connected to the failure of the cells to enter S phase due to apoptosis in these cells. The inhibition of growth may be due to the triggering of apoptosis and inhibition of cell cycle at G1 phase (Fig. 5c, 5d). It is well known that most of the available antifungal's act through the production of ROS this helps them in their fungicidal actions. It was previously reported that by blocking the mitochondrial electron transport chain, farnesol, a quorum-sensing molecule in *C. albicans*, may cause the production of ROS [36]. Therefore, we studied the effect of zingerone on ROS production and it was found that zingerone induces ROS accumulation in *C. albicans* ATCC 90028 and C1 isolate (Fig. 6a).

The present study investigates the inhibitory effects of zingerone on the morphogenic switch and biofilm formation in *C. albicans*, providing insights into the underlying molecular mechanisms. The qRT-PCR results reveal significant alterations in the expression of key genes involved in hyphal development and signaling pathways, highlighting the strain-specific responses to zingerone treatment. In *C. albicans* ATCC 90028, the downregulation of negative regulators *NRG1* and *MIG1* by 1.07-fold and 1.2-fold, respectively, suggests a potential pathway for increased filamentous growth. However, the response is quite different in the C1 clinical isolate, where *NRG1* and *MIG1* were markedly upregulated by 4.3-fold and 10.84-fold. This differential expression indicates that zingerone may interact with distinct regulatory networks in different strains, affecting the balance between yeast and hyphal forms.

The upregulation of *TUP1* in both strains 2.19-fold in *C. albicans* ATCC 90028 and 28.7-fold in the C1 isolate further supports zingerone's role as a modulator of hyphal growth. *TUP1* acts as a repressor of hyphal development, and its enhanced expression suggests a potential mechanism through which zingerone limits biofilm formation. The contrasting expression patterns of *BCY1*, which was downregulated in ATCC 90028 but upregulated in the C1 isolate, further emphasize the complexity of the signaling pathways involved and the possible strain-specific regulatory responses to zingerone. The analysis of the Ras1-cAMP-Efg1 and Cek1-MAPK pathways reveals substantial downregulation of key genes associated with pathogenicity. The significant reduction in *RAS1* expression by 6.27-fold in

C. albicans ATCC 90028 and 5.89-fold in the C1 isolate underscores the importance of this pathway in *C. albicans* pathogenicity and suggests that zingerone may impair the organism's ability to form biofilms through this mechanism. Similarly, *PDE2*, which is crucial for maintaining hyphal growth and cell wall integrity, was downregulated, with a greater reduction observed in the C1 isolate.

Moreover, the consistent downregulation of *EFG1* and *TEC1* across both strains reinforces the notion that zingerone disrupts signaling pathways essential for biofilm development. The reductions of 12.47-fold and 49.84-fold in *EFG1* expression, along with the downregulation of downstream genes such as *ECE1* and *HWPI*, indicate a broad suppression of factors promoting hyphal formation. The significant downregulation of these genes in both strains suggests a strong inhibitory effect of zingerone on biofilm formation, which is crucial for *C. albicans* pathogenicity.

In the Cek1-MAPK pathway, the downregulation of *CPH1* and *HST7* indicates a further suppression of hyphal development. The substantial reductions in *CEK1* expression 21.42-fold in *C. albicans* ATCC 90028 and 2.48-fold in C1 isolate reflect the potential impact of zingerone on pathways regulating morphogenesis.

Additionally, the expression patterns of *NRG1* and *MIG1* in the context of the two strains provide intriguing insights. The downregulation of *NRG1* and *MIG1* in ATCC 90028 suggests a potential for hyphal development that is not realized due to the overall inhibitory effects of zingerone. In contrast, the C1 isolate's differential expression downregulation of *NRG1* and upregulation of *MIG1* highlights a unique adaptive response that warrants further investigation.

Hence zingerone exhibits significant inhibitory effects on the morphogenic switch and biofilm formation in *C. albicans*, mediated through complex regulatory mechanisms involving key signaling pathways. These findings underscore the potential of zingerone as a novel therapeutic agent in the management of *C. albicans* infections, particularly in strains exhibiting increased resistance to conventional antifungal treatments. Further studies are warranted to explore the clinical implications of these findings and the potential for zingerone in antifungal drug development.

This study further delves into the effects of zingerone on the expression of genes related to oxidative stress protection, providing insight into its impact on reactive oxygen species (ROS) production and cell cycle dynamics in *C. albicans*. The examination of genes such as *SOD1*, *SOD2*, *CAP1*, *CAT1*, and *MCA1* reveals distinct expression patterns that underscore the varying responses of different *C. albicans* strains to zingerone treatment.

In *C. albicans* ATCC 90028, the observed reductions in *SOD2* by 1.18-fold and *CAT1* by 1.67-fold expression suggest a decreased capacity for oxidative stress defense following zingerone treatment. This downregulation could lead to an accumulation of ROS, which may induce cellular stress and impact the cell cycle. Conversely, in the C1 clinical isolate, *SOD2* was downregulated by 1.30-fold, while *CAT1* was notably upregulated by 1.67-fold. This differential response indicates that the C1 isolate may activate certain protective mechanisms in response to zingerone, potentially to counteract the oxidative stress induced by ROS production.

The significant overexpression of *SOD1* in the C1 isolate post-treatment further emphasizes its role in the organism's defense against oxidative stress. This gene is crucial for scavenging superoxide radicals, and its upregulation may help mitigate the detrimental effects of increased ROS levels. Such a response might suggest an adaptive mechanism that allows the C1 isolate to better withstand the oxidative environment created by zingerone treatment.

Interestingly, the expression of *MCA1*, a gene associated with apoptosis, exhibited contrasting trends between the two strains. In *C. albicans* ATCC 90028, *MCA1* was upregulated by 1.38-fold, indicating a potential activation of apoptotic pathways in response to zingerone exposure. This increase may correlate with heightened ROS levels and cellular stress, prompting a programmed cell death

response. In contrast, the downregulation of *MCA1* by 1.54-fold in the C1 isolate suggests a different strategy, possibly indicating a shift away from apoptosis in favor of survival mechanisms in the face of zingerone treatment.

These distinct expression patterns reveal how zingerone differentially impacts oxidative stress responses and apoptosis in *C. albicans*, highlighting the complexity of its effects based on the strain. The observed changes in gene expression are likely to contribute to variations in ROS generation and subsequent cell cycle dynamics, providing deeper insight into the molecular mechanisms through which zingerone affects *C. albicans*.

The use of acridine orange (AO) and ethidium bromide (EB) in dual staining effectively assessed cell viability in *C. albicans* after zingerone treatment. In the control group, a high density of live cells was observed, indicated by green fluorescence. However, post-treatment with zingerone, *C. albicans* ATCC 90028 showed a significant shift to red fluorescence, indicating cell death, which highlights zingerone's effectiveness in inducing cytotoxicity in this strain. In contrast, the C1 isolate retained a significant number of live cells, as evidenced by the green fluorescence, suggesting that zingerone did not have a lethal effect on this strain. This differential response underscores the strain-specific mechanisms of action and highlights the need for further investigation into the molecular factors contributing to this variability.

In addition to all these activities zingerone was not showing any significant hemolysis up to 4 mg/ml concentration (Fig. 7b). Also, previously it was reported that the silk worms get killed by pathogenic fungi [25]. Within this study it was demonstrated that *C. albicans* infection killed the silk worm within 24 h time interval but when the MIC concentration of zingerone was given to the silk worm the survival time increased up to 48 to 72 h even after the infection and went into the pupation stage and complete normal life cycle (Fig. 7). Hence it was confirmed that zingerone increased the survival of silk worm in both the cases.

The study has several limitations. Firstly, the relatively small sample size may impact the generalizability of the results, suggesting that larger sample sizes in future research could yield more robust and reliable data. Additionally, the investigation primarily focused on a single *C. albicans* strain and a clinical isolate, indicating a need for broader strain testing to gain a more comprehensive understanding of zingerone's effects across different species. The use of the silkworm model, while informative, presents challenges in translating the findings directly to human physiology; therefore, studies utilizing mammalian models could provide deeper insights into the efficacy and safety of zingerone.

Despite these limitations, the study provides valuable insights into the anti-virulence effects of zingerone on *C. albicans*, contributing to the understanding of potential therapeutic agents against fungal infections.

Conclusions

The findings of this study highlight the potent antifungal properties of zingerone against *Candida albicans*, showcasing its ability to inhibit planktonic growth, disrupt biofilm formation, and impair morphological transitions. Zingerone demonstrated a concentration-dependent inhibition of growth, with significant reductions in viability observed for the standard strain *C. albicans* ATCC 90028, achieving a MIC of 2 mg/mL and a complete eradication at 4 mg/mL. Conversely, the clinical isolate C1 required higher concentrations for similar effects, indicating strain-specific responses.

Zingerone effectively inhibited the adhesion of *C. albicans* to surfaces, crucial for biofilm formation, achieving up to 97% reduction at higher concentrations. The compound also significantly disrupted the yeast-to-hyphal transition, which is critical for *C. albicans* pathogenicity, with complete inhibition at 4 mg/mL. Qualitative and

quantitative assessments of biofilm formation revealed that zingerone effectively reduced biofilm density and thickness, further substantiating its role as a biofilm inhibitor.

Additionally, zingerone induced cell membrane damage, evidenced by increased propidium iodide uptake and changes in ergosterol biosynthesis, indicating a severe impact on membrane integrity. Flow cytometry analyses revealed zingerone-induced cell cycle arrest and apoptosis, alongside alterations in gene expression related to ROS production, morphogenesis, and biofilm development.

Despite its efficacy, zingerone exhibited negligible hemolytic activity, suggesting a favorable safety profile for potential therapeutic applications. Overall, zingerone emerges as a promising candidate for the development of antifungal therapies targeting *C. albicans*, warranting further investigation into its mechanisms of action and potential clinical applications. Preclinical studies in animals need to be performed to confirm antifungal efficacy of zingerone.

Availability of data and materials

The dataset analyzed during our current study is available from the corresponding author on reasonable request.

Ethical approval

Not applicable.

Consent for publication

Not applicable.

Declaration of competing interest

The authors involved in this work declare that they have no conflicts of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.mycmed.2024.101527](https://doi.org/10.1016/j.mycmed.2024.101527).

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